```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib
import plotly.graph_objects as go
import plotly.express as px
from plotly.subplots import make_subplots
import plotly.io as pio
import seaborn as sns

%matplotlib inline

pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', 500)
pd.set_option('display.expand_frame_repr', False)
```

Dataset Exploration

```python
df = pd.read_csv("../data/raw/UNSW_NB15_training-set.csv")
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 82332 entries, 0 to 82331
Data columns (total 45 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   id                 82332 non-null  int64
 1   dur                82332 non-null  float64
 2   proto              82332 non-null  str
 3   service            82332 non-null  str
 4   state              82332 non-null  str
 5   spkts              82332 non-null  int64
 6   dpkts              82332 non-null  int64
 7   sbytes             82332 non-null  int64
 8   dbytes             82332 non-null  int64
 9   rate               82332 non-null  float64
 10  sttl               82332 non-null  int64
 11  dttl               82332 non-null  int64
 12  sload              82332 non-null  float64
 13  dload              82332 non-null  float64
 14  sloss              82332 non-null  int64
 15  dloss              82332 non-null  int64
 16  sinpkt             82332 non-null  float64
 17  dinpkt             82332 non-null  float64
 18  sjit               82332 non-null  float64
 19  djit               82332 non-null  float64
 20  swin               82332 non-null  int64
 21  stcpb              82332 non-null  int64
```

```
 22  dtcpb               82332 non-null  int64
 23  dwin                82332 non-null  int64
 24  tcprtt              82332 non-null  float64
 25  synack              82332 non-null  float64
 26  ackdat              82332 non-null  float64
 27  smean               82332 non-null  int64
 28  dmean               82332 non-null  int64
 29  trans_depth         82332 non-null  int64
 30  response_body_len   82332 non-null  int64
 31  ct_srv_src          82332 non-null  int64
 32  ct_state_ttl        82332 non-null  int64
 33  ct_dst_ltm          82332 non-null  int64
 34  ct_src_dport_ltm    82332 non-null  int64
 35  ct_dst_sport_ltm    82332 non-null  int64
 36  ct_dst_src_ltm      82332 non-null  int64
 37  is_ftp_login        82332 non-null  int64
 38  ct_ftp_cmd          82332 non-null  int64
 39  ct_flw_http_mthd    82332 non-null  int64
 40  ct_src_ltm          82332 non-null  int64
 41  ct_srv_dst          82332 non-null  int64
 42  is_sm_ips_ports     82332 non-null  int64
 43  attack_cat          82332 non-null  str
 44  label               82332 non-null  int64
dtypes: float64(11), int64(30), str(4)
memory usage: 28.3 MB
```

```
df
```

```
          id       dur proto service state  spkts  dpkts  sbytes
dbytes           rate  sttl  dttl         sload        dload  sloss
dloss        sinpkt  dinpkt         sjit        djit  swin       stcpb
dtcpb  dwin    tcprtt    synack    ackdat  smean  dmean  trans_depth
response_body_len  ct_srv_src  ct_state_ttl  ct_dst_ltm
ct_src_dport_ltm  ct_dst_sport_ltm  ct_dst_src_ltm  is_ftp_login
ct_ftp_cmd  ct_flw_http_mthd  ct_src_ltm  ct_srv_dst  is_sm_ips_ports
attack_cat  label
0          1  0.000011   udp       -   INT      2      0     496
0   90909.090200   254     0  1.803636e+08     0.000000      0      0
0.011000     0.0     0.000000    0.000000     0           0
0     0  0.000000  0.000000  0.000000    248      0            0
0           2             2           1                 1
1               2             0           0                 0
1           2                0     Normal      0
1          2  0.000008   udp       -   INT      2      0    1762
0  125000.000300   254     0  8.810000e+08     0.000000      0      0
0.008000     0.0     0.000000    0.000000     0           0
0     0  0.000000  0.000000  0.000000    881      0            0
0           2             2           1                 1
1               2             0           0                 0
1           2                0     Normal      0
```

```
2          3  0.000005   udp       -   INT      2      0    1068
0  200000.005100   254     0  8.544000e+08     0.000000      0      0
0.005000     0.0     0.000000    0.000000     0           0
0     0  0.000000  0.000000  0.000000    534      0            0
0           3             2           1                 1
1               3             0           0                 0
1           3                0     Normal      0
3          4  0.000006   udp       -   INT      2      0     900
0  166666.660800   254     0  6.000000e+08     0.000000      0      0
0.006000     0.0     0.000000    0.000000     0           0
0     0  0.000000  0.000000  0.000000    450      0            0
0           3             2           2                 2
1               3             0           0                 0
2           3                0     Normal      0
4          5  0.000010   udp       -   INT      2      0    2126
0  100000.002500   254     0  8.504000e+08     0.000000      0      0
0.010000     0.0     0.000000    0.000000     0           0
0     0  0.000000  0.000000  0.000000   1063      0            0
0           3             2           2                 2
1               3             0           0                 0
2           3                0     Normal      0
...      ...       ...   ...     ...   ...    ...    ...     ...     .
..            ...   ...   ...           ...          ...    ...    ...
...     ...          ...         ...   ...         ...         ...   .
..       ...       ...       ...    ...    ...          ...
...         ...           ...         ...               ...
...             ...           ...         ...                ...
...          ...               ...        ...    ...
82327  82328  0.000005   udp       -   INT      2      0     104
0  200000.005100   254     0  8.320000e+07     0.000000      0      0
0.005000     0.0     0.000000    0.000000     0           0
0     0  0.000000  0.000000  0.000000     52      0            0
0           1             2           2                 1
1               2             0           0                 0
2           1                0     Normal      0
82328  82329  1.106101   tcp       -   FIN     20      8   18062
354      24.410067   254   252  1.241044e+05  2242.109863      7
1     55.880051   143.7  4798.130981  190.980813   255  1072535109
3284291478   255  0.173208  0.100191  0.073017    903     44
0                 0           1             1           2
1                 1               1             0           0
0           3           2                0     Normal      0
82329  82330  0.000000   arp       -   INT      1      0      46
0       0.000000     0     0  0.000000e+00     0.000000      0      0
60000.720000     0.0     0.000000    0.000000     0           0
0     0  0.000000  0.000000  0.000000     46      0            0
0           1             2           1                 1
1               1             0           0                 0
1           1                1     Normal      0
```

```
82330  82331  0.000000   arp       -   INT      1      0      46
0       0.000000     0     0  0.000000e+00     0.000000      0      0
60000.732000     0.0    10.954518    0.000000     0           0
0     0  0.000000  0.000000  0.000000     46      0            0
0           1             2           1                 1
1               1             0           0                 0
1           1                1     Normal      0
82331  82332  0.000009   udp       -   INT      2      0     104
0  111111.107200   254     0  4.622222e+07     0.000000      0      0
0.009000     0.0     0.000000    0.000000     0           0
0     0  0.000000  0.000000  0.000000     52      0            0
0           1             2           1                 1
1               1             0           0                 0
1           1                0     Normal      0

[82332 rows x 45 columns]
```

First check for duplicates

```
n_before = len(df)
df = df.drop_duplicates(subset=[c for c in df.columns if c !=
'id']).reset_index(drop=True)
print(f"Duplicate rows removed: {n_before - len(df)}")
print(f"Remaining rows: {len(df)}")
```

```
Duplicate rows removed: 26387
Remaining rows: 55945
```

```
df['label'].value_counts()
```

```
label
0    34206
1    21739
Name: count, dtype: int64
```

This dataset is almost balanced, with 51.8% of training data being labelled as attack (1) vs 48.2% being labelled as normal (0).

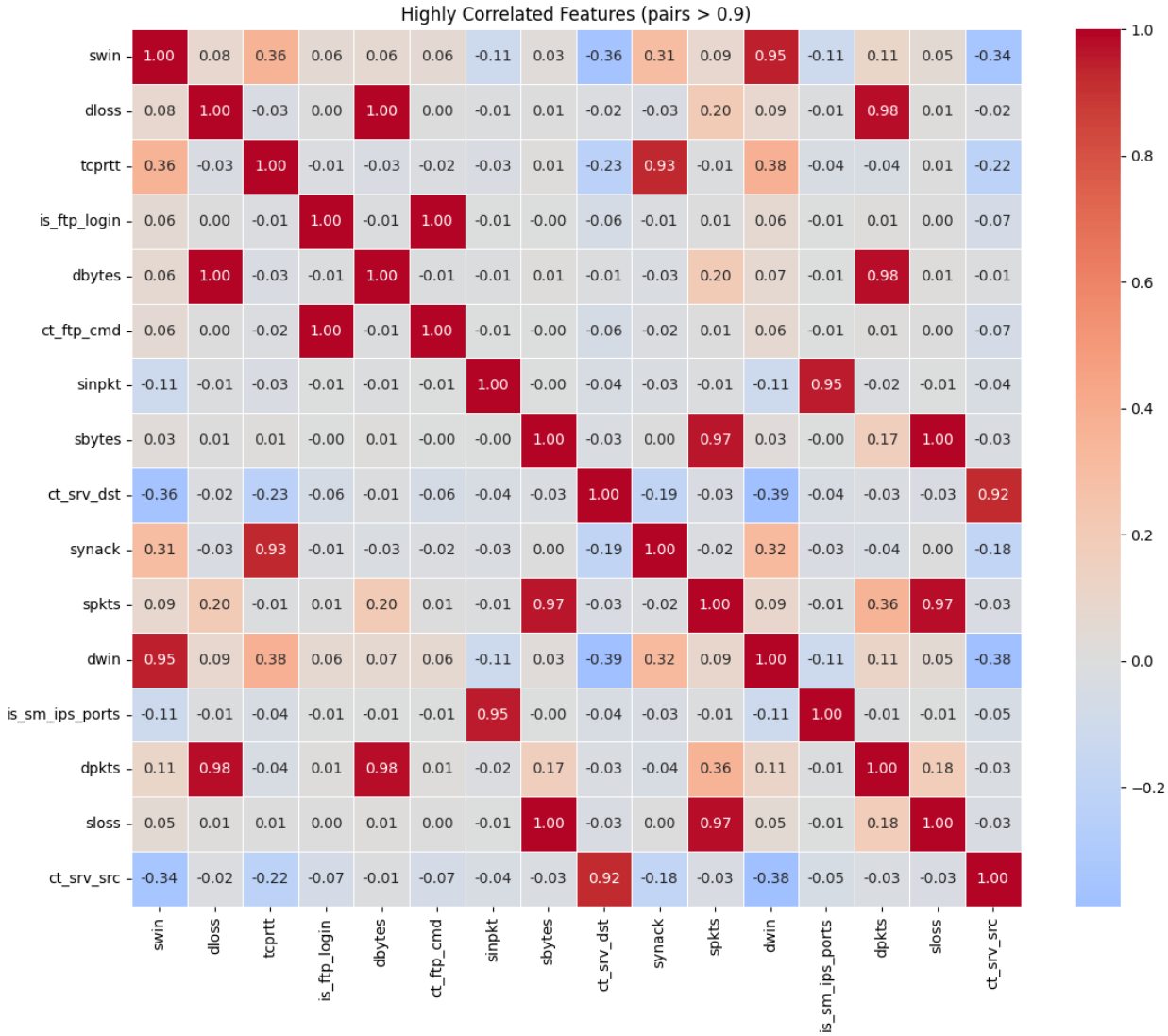
Correlation Heatmap Analysis, excluding irrelevant columns (id, label, stcpb, dtcpb)

```
num_cols_eda =
df.select_dtypes(include='number').columns.difference(['id', 'label',
'stcpb', 'dtcpb']).tolist()

corr_matrix = df[num_cols_eda].corr().abs()
upper = corr_matrix.where(np.triu(np.ones(corr_matrix.shape),
k=1).astype(bool))
high_corr = [(col, row, round(upper.loc[row, col], 3))
             for col in upper.columns for row in upper.index
             if pd.notna(upper.loc[row, col]) and upper.loc[row, col]
```

```
> 0.9]
high_corr.sort(key=lambda x: x[2], reverse=True)

print(f"Feature pairs with correlation > 0.9: {len(high_corr)}")
for a, b, v in high_corr:
    print(f"  {a:30s} <-> {b:30s}  {v}")
```

```
Feature pairs with correlation > 0.9: 11
  dloss                          <-> dbytes
0.997
  sloss                          <-> sbytes
0.996
  is_ftp_login                   <-> ct_ftp_cmd
0.995
  dpkts                          <-> dloss
0.983
  dpkts                          <-> dbytes
0.978
  spkts                          <-> sloss
0.974
  spkts                          <-> sbytes
0.966
  sinpkt                         <-> is_sm_ips_ports
0.954
  swin                           <-> dwin
0.95
  tcprtt                         <-> synack
0.935
  ct_srv_src                     <-> ct_srv_dst
0.924
```

```
if high_corr:
    corr_features = list(set([f for a, b, _ in high_corr for f in (a,
b)]))
    fig, ax = plt.subplots(figsize=(12, 10))
    sns.heatmap(df[corr_features].corr(), annot=True, fmt='.2f',
cmap='coolwarm',
                center=0, ax=ax, linewidths=0.5)
    ax.set_title('Highly Correlated Features (pairs > 0.9)')
    plt.tight_layout()
    plt.show()
else:
    print("No feature pairs with correlation > 0.9")
```



Feature Engineering

Let's conduct an EDA on our features to determine any pre-processing that needs to be done within our dataset, as well as if there are any features to exclude or modify to train our models.

First, drop the irrelevant columns being 'id' and 'attack_cat' as we do not need the id nor are we classifying the attack type (a different problem). Additionally, drop the columns 'stcpb' and 'dtcpb' as they are random TCP sequence numbers and will not contribute to the predictions.

```
df.drop(['id', 'attack_cat', 'stcpb', 'dtcpb'],axis=1,inplace=True)
```

Now get categorical vs numerical features

```
cat_cols = df.select_dtypes(include=['str']).columns.tolist()
num_cols =
df.select_dtypes(include='number').columns.difference(['label']).tolis
```

```
t()

print(f"Categorical features ({len(cat_cols)}): {cat_cols}")
print(f"Numerical features  ({len(num_cols)}): {num_cols}")
```

```
Categorical features (3): ['proto', 'service', 'state']
Numerical features  (37): ['ackdat', 'ct_dst_ltm', 'ct_dst_sport_ltm',
'ct_dst_src_ltm', 'ct_flw_http_mthd', 'ct_ftp_cmd',
'ct_src_dport_ltm', 'ct_src_ltm', 'ct_srv_dst', 'ct_srv_src',
'ct_state_ttl', 'dbytes', 'dinpkt', 'djit', 'dload', 'dloss', 'dmean',
'dpkts', 'dttl', 'dur', 'dwin', 'is_ftp_login', 'is_sm_ips_ports',
'rate', 'response_body_len', 'sbytes', 'sinpkt', 'sjit', 'sload',
'sloss', 'smean', 'spkts', 'sttl', 'swin', 'synack', 'tcprtt',
'trans_depth']
```

Categorical Features — Unique counts & distributions

```
for col in cat_cols:
    vc = df[col].value_counts()
    print(f"\n{col}  ({vc.shape[0]} unique values)")
    print(vc.to_string())
```

```
proto  (131 unique values)
proto
tcp            40559
udp            12541
unas             885
arp              298
ospf             190
sctp              77
any               30
igmp              26
gre               21
rsvp              20
sun-nd            17
emcon             16
nvp               16
ipv6              16
mobile            15
vmtp              15
swipe             15
sep               15
secure-vmtp       15
pim               15
snp               15
prm               12
xns-idp           12
leaf-1            12
bbn-rcc           12
```

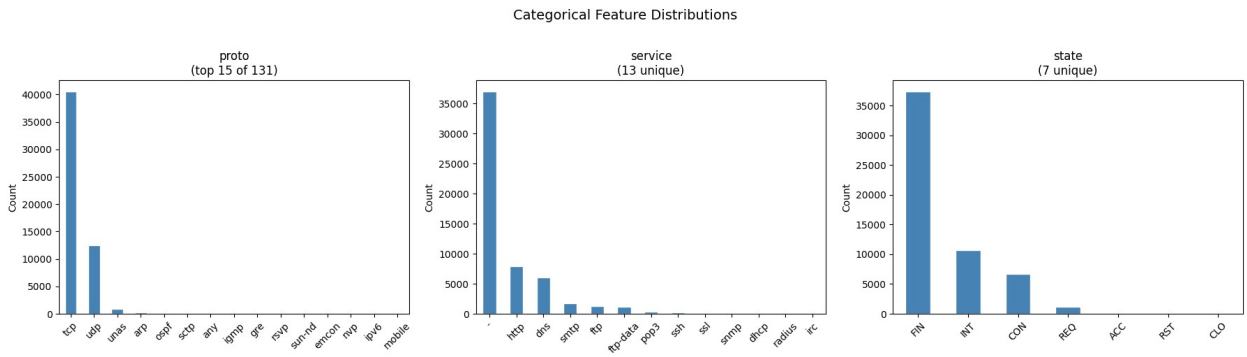
```
ggp                 11
ip                  11
ipnip               11
st2                 11
argus               11
chaos               11
egp                 11
pup                 11
xnet                11
mux                 11
dcn                 11
hmp                 11
trunk-1             11
trunk-2             11
leaf-2              11
irtp                11
rdp                 11
netblt              11
cbt                 11
igp                 11
iso-tp4             11
mfe-nsp             10
merit-inp           10
3pc                 10
idpr                10
ddp                 10
idpr-cmtp           10
tp++                10
sdrp                10
ipv6-frag           10
ipv6-route          10
idrp                10
mhrp                10
i-nlsp              10
rvd                 10
narp                10
skip                10
tlsp                10
ipv6-no             10
ipv6-opts           10
cftp                10
sat-expak           10
ippc                10
kryptolan           10
sat-mon             10
cpnx                10
wsn                 10
pvp                 10
br-sat-mon          10
```

```
wb-mon              10
ttp                 10
vines               10
nsfnet-igp          10
dgp                 10
eigrp               10
tcf                 10
sprite-rpc          10
larp                10
mtp                 10
ax.25               10
ipip                10
aes-sp3-d           10
micp                10
encap               10
pri-enc             10
gmtp                10
ifmp                10
pnni                10
qnx                 10
scps                10
bna                 10
visa                10
ipcv                10
cphb                10
wb-expak            10
xtp                 10
il                  10
fc                  10
iso-ip              10
etherip             10
aris                10
a/n                 10
ipcomp              10
compaq-peer         10
ipx-n-ip            10
pgm                 10
vrrp                10
l2tp                10
zero                10
ddx                 10
iatp                10
stp                 10
srp                 10
uti                 10
sm                  10
smp                 10
isis                10
ptp                 10
```

```
fire                10
crtp                10
crudp               10
sccopmce            10
iplt                10
pipe                10
sps                 10
ib                   9

service  (13 unique values)
service
-            37027
http          7930
dns           6091
smtp          1759
ftp           1291
ftp-data      1177
pop3           381
ssh            204
ssl             30
snmp            25
dhcp            19
radius           6
irc              5

state  (7 unique values)
state
FIN    37355
INT    10646
CON     6711
REQ     1227
ACC        4
RST        1
CLO        1
```

```
fig, axes = plt.subplots(1, len(cat_cols), figsize=(6 * len(cat_cols),
5))

for ax, col in zip(axes, cat_cols):
    vc = df[col].value_counts()
    # Cap at top 15 as proto has 131 values
    if len(vc) > 15:
        vc = vc.head(15)
        ax.set_title(f'{col}\n(top 15 of {df[col].nunique()})')
    else:
        ax.set_title(f'{col}\n({len(vc)} unique)')
    vc.plot(kind='bar', ax=ax, color='steelblue', edgecolor='white')
    ax.set_xlabel('')
    ax.set_ylabel('Count')
    ax.tick_params(axis='x', rotation=45)
```

```
plt.suptitle('Categorical Feature Distributions', fontsize=14, y=1.02)
plt.tight_layout()
plt.show()
```



Numerical Features

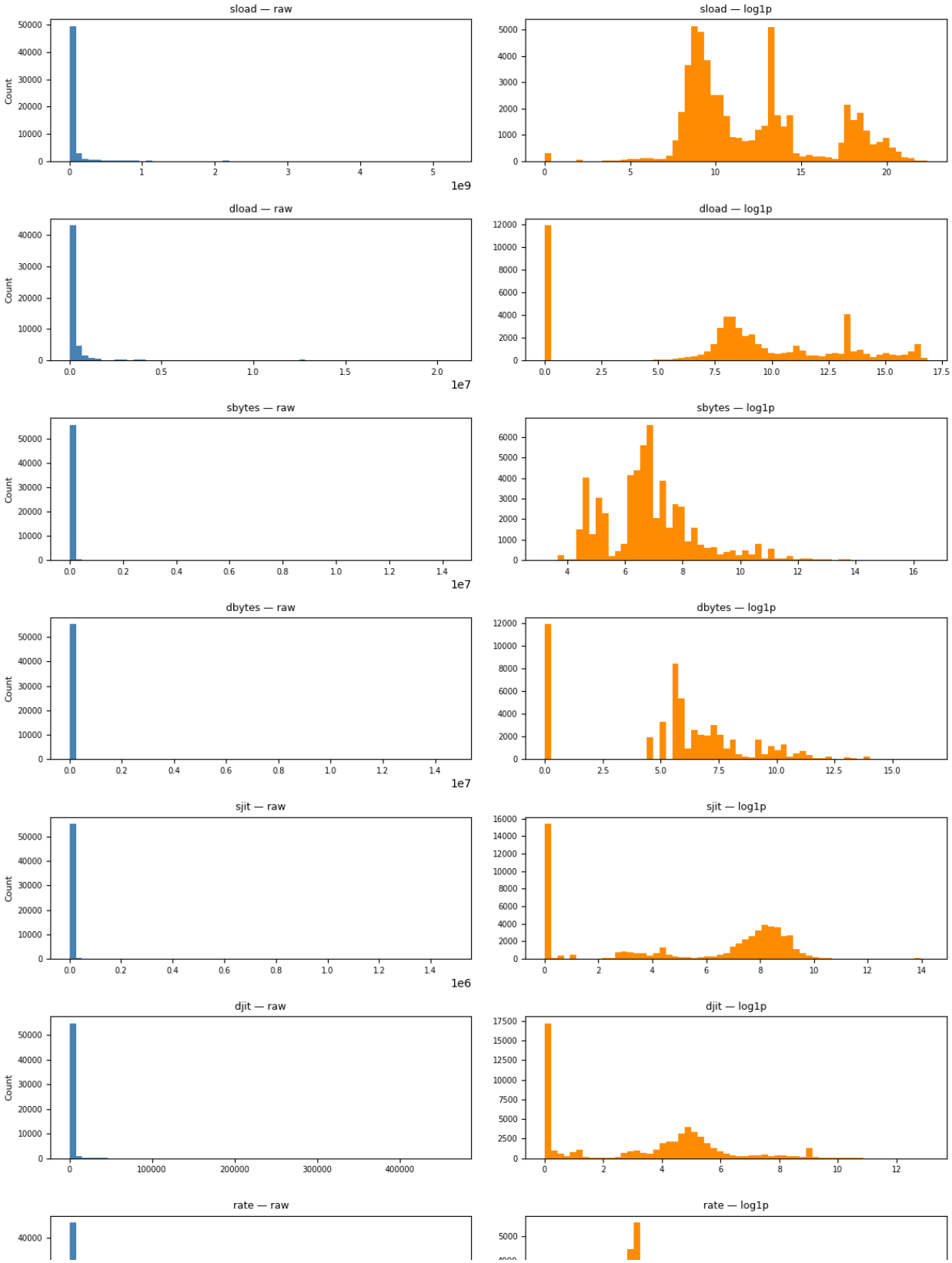
```
stats = df[num_cols].describe().T[['min', 'max', 'mean', 'std']]
stats['range'] = stats['max'] - stats['min']
stats.sort_values('range', ascending=False).style.format({
    'min':   '{:,.2f}',
    'max':   '{:,.2f}',
    'mean':  '{:,.2f}',
    'std':   '{:,.2f}',
    'range': '{:,.2f}',
})
```

```
<pandas.io.formats.style.Styler at 0x1334a9110>
```

```
skewed_cols = [
    'sload', 'dload', 'sbytes', 'dbytes', 'sjit', 'djit',
    'rate', 'sinpkt', 'dinpkt', 'spkts', 'dpkts',
    'sloss', 'dloss', 'response_body_len'
]

n = len(skewed_cols)
fig, axes = plt.subplots(n, 2, figsize=(12, n * 2.5))

for i, col in enumerate(skewed_cols):
    raw = df[col].dropna()
    log = np.log1p(raw)

    axes[i, 0].hist(raw, bins=60, color='steelblue', edgecolor='none')
    axes[i, 0].set_title(f'{col} — raw', fontsize=9)
    axes[i, 0].set_ylabel('Count', fontsize=8)
    axes[i, 0].tick_params(labelsize=7)

    axes[i, 1].hist(log, bins=60, color='darkorange',
edgecolor='none')
```

```
    axes[i, 1].set_title(f'{col} — log1p', fontsize=9)
    axes[i, 1].tick_params(labelsize=7)

plt.suptitle('Skewed Features: Raw vs log1p Distributions',
fontsize=12, y=1.01)
plt.tight_layout()
plt.show()
```

Skewed Features: Raw vs log1p Distributions



Preprocessing

Categorical Variables

We conduct preprocessing based on the results of the EDA above. There are three categorical variables: proto, service, and state. For proto, we can see that only 6 out of its 131 unique categories/values have counts of at least 100 in our dataset of 82 322. To prevent falling under the curse of dimensionality and unnessecary complexity when training our different models, we have made the choice to combine all other categories under one new category named "other", to reduce our unique values from 131 to 7.

This is the same for the service column, where ...

And for state ..

```
from sklearn.preprocessing import StandardScaler

proto_filter   = ['tcp', 'udp', 'unas', 'arp', 'ospf', 'sctp']
service_filter = ['-', 'dns', 'http', 'smtp', 'ftp', 'ftp-data',
'ssh', 'pop3']
state_filter  = ['FIN', 'INT', 'CON', 'REQ']

df['proto_filtered']   =
df['proto'].where(df['proto'].isin(proto_filter), other='other')
df['service_filtered'] =
df['service'].where(df['service'].isin(service_filter), other='other')
df['state_filtered']   =
df['state'].where(df['state'].isin(state_filter), other='other')

proto_ohe   = pd.get_dummies(df['proto_filtered'],   prefix='proto')
service_ohe = pd.get_dummies(df['service_filtered'], prefix='service')
state_ohe   = pd.get_dummies(df['state_filtered'],   prefix='state')

ohe_cols = proto_ohe.columns.tolist() + service_ohe.columns.tolist() +
state_ohe.columns.tolist()

df = pd.concat([df, proto_ohe, service_ohe, state_ohe], axis=1)

df.columns
```

```
Index(['dur', 'proto', 'service', 'state', 'spkts', 'dpkts', 'sbytes',
       'dbytes', 'rate', 'sttl', 'dttl', 'sload', 'dload', 'sloss',
'dloss',
       'sinpkt', 'dinpkt', 'sjit', 'djit', 'swin', 'dwin', 'tcprtt',
'synack',
       'ackdat', 'smean', 'dmean', 'trans_depth', 'response_body_len',
       'ct_srv_src', 'ct_state_ttl', 'ct_dst_ltm', 'ct_src_dport_ltm',
       'ct_dst_sport_ltm', 'ct_dst_src_ltm', 'is_ftp_login',
'ct_ftp_cmd',
       'ct_flw_http_mthd', 'ct_src_ltm', 'ct_srv_dst',
```

```
'is_sm_ips_ports',
       'label', 'proto_filtered', 'service_filtered',
'state_filtered',
       'proto_arp', 'proto_ospf', 'proto_other', 'proto_sctp',
'proto_tcp',
       'proto_udp', 'proto_unas', 'service_-', 'service_dns',
'service_ftp',
       'service_ftp-data', 'service_http', 'service_other',
'service_pop3',
       'service_smtp', 'service_ssh', 'state_CON', 'state_FIN',
'state_INT',
       'state_REQ', 'state_other'],
      dtype='str')
```

Numerical Features

Apply StandardScaler to all numerical columns, with scaled columns being added with a _scaled suffix so originals are preserved.

Additionally, there are multiple columns which are right skewed, as shown from the numerical chart and skewed column plots we displayed earlier. We use a log transformation on these columns to handle the skewness so the distribution appears more normal.

```
skewed_cols = [
    'sload', 'dload', 'sbytes', 'dbytes', 'sjit', 'djit',
    'rate', 'sinpkt', 'dinpkt', 'spkts', 'dpkts',
    'sloss', 'dloss', 'response_body_len'
]
non_skewed_cols = [c for c in num_cols if c not in skewed_cols]

for col in skewed_cols:
    df[f'{col}_log'] = np.log1p(df[col])

log_cols = [f'{col}_log' for col in skewed_cols]

scaler = StandardScaler()
cols_to_scale = log_cols + non_skewed_cols
scaled_values = scaler.fit_transform(df[cols_to_scale])
scaled_cols = [f'{c}_scaled' for c in cols_to_scale]
df = pd.concat([df, pd.DataFrame(scaled_values, columns=scaled_cols,
index=df.index)], axis=1)

df
```

```
            dur proto service state  spkts  dpkts  sbytes  dbytes
rate  sttl  dttl         sload        dload  sloss  dloss     sinpkt
dinpkt         sjit        djit  swin  dwin    tcprtt    synack
ackdat  smean  dmean  trans_depth  response_body_len  ct_srv_src
ct_state_ttl  ct_dst_ltm  ct_src_dport_ltm  ct_dst_sport_ltm
ct_dst_src_ltm  is_ftp_login  ct_ftp_cmd  ct_flw_http_mthd  ct_src_ltm
```

```
ct_srv_dst  is_sm_ips_ports  label proto_filtered service_filtered
state_filtered  proto_arp  proto_ospf  proto_other  proto_sctp
proto_tcp  proto_udp  proto_unas  service_-  service_dns  service_ftp
service_ftp-data  service_http  service_other  service_pop3
service_smtp  service_ssh  state_CON  state_FIN  state_INT  state_REQ
state_other  sload_log  dload_log  sbytes_log  dbytes_log  sjit_log
djit_log   rate_log  sinpkt_log  dinpkt_log  spkts_log  dpkts_log
sloss_log  dloss_log  response_body_len_log  sload_log_scaled
dload_log_scaled  sbytes_log_scaled  dbytes_log_scaled
sjit_log_scaled  djit_log_scaled  rate_log_scaled  sinpkt_log_scaled
dinpkt_log_scaled  spkts_log_scaled  dpkts_log_scaled
sloss_log_scaled  dloss_log_scaled  response_body_len_log_scaled
ackdat_scaled  ct_dst_ltm_scaled  ct_dst_sport_ltm_scaled
ct_dst_src_ltm_scaled  ct_flw_http_mthd_scaled  ct_ftp_cmd_scaled
ct_src_dport_ltm_scaled  ct_src_ltm_scaled  ct_srv_dst_scaled
ct_srv_src_scaled  ct_state_ttl_scaled  dmean_scaled  dttl_scaled
dur_scaled  dwin_scaled  is_ftp_login_scaled  is_sm_ips_ports_scaled
smean_scaled  sttl_scaled  swin_scaled  synack_scaled  tcprtt_scaled
trans_depth_scaled
0      0.000011   udp       -   INT      2      0     496       0
90909.090200   254     0  1.803636e+08     0.000000      0      0
0.011000    0.000000     0.000000    0.000000     0     0  0.000000
0.000000  0.000000    248      0            0                  0
2             2           1                 1                 1
2             0           0                 0           1           2
0      0            udp                -            INT      False
False        False       False      False       True       False
True        False        False             False         False
False         False         False        False      False      False
True      False        False  19.010486   0.000000    6.208590
0.000000  0.000000  0.000000  11.417626    0.010940    0.000000
1.098612   0.000000   0.000000   0.000000                    0.0
1.761786         -1.649765          -0.419856          -1.650266
-1.405884        -1.200567         1.516892          -1.185791
-1.103353         -1.217874         -1.440439         -1.199338
-0.956103                      -0.307674      -0.590835          -
0.503144                -0.217036              -0.285107
-0.280678          -0.097247                -0.311272          -
0.588983          -0.516027          -0.574568             0.857978
-0.591384    -1.116899   -0.252515    -1.542394            -0.097331
-0.069379      0.340845     0.933831    -1.623348      -0.496312
-0.588361           -0.203984
1      0.000008   udp       -   INT      2      0    1762       0
125000.000300   254     0  8.810000e+08     0.000000      0      0
0.008000    0.000000     0.000000    0.000000     0     0  0.000000
0.000000  0.000000    881      0            0                  0
2             2           1                 1                 1
2             0           0                 0           1           2
0      0            udp                -            INT      False
```

False False False False True False
True False False False False
False False False False False False
True False False 20.596568 0.000000 7.474772
0.000000 0.000000 0.000000 11.736077 0.007968 0.000000
1.098612 0.000000 0.000000 0.000000 0.0
2.168308 -1.649765 0.387922 -1.650266
-1.405884 -1.200567 1.606793 -1.187154
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.285107
-0.280678 -0.097247 -0.311272 -
0.588983 -0.516027 -0.574568 0.857978
-0.591384 -1.116899 -0.252515 -1.542394 -0.097331
-0.069379 2.986899 0.933831 -1.623348 -0.496312
-0.588361 -0.203984
2 0.000005 udp - INT 2 0 1068 0
200000.005100 254 0 8.544000e+08 0.000000 0 0
0.005000 0.000000 0.000000 0.000000 0 0 0.000000
0.000000 0.000000 534 0 0 0
3 2 1 1 1
3 0 0 0 1 3
0 0 udp - INT False
False False False False True False
True False False False False
False False False False False False
True False False 20.565910 0.000000 6.974479
0.000000 0.000000 0.000000 12.206078 0.004988 0.000000
1.098612 0.000000 0.000000 0.000000 0.0
2.160450 -1.649765 0.068753 -1.650266
-1.405884 -1.200567 1.739477 -1.188520
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.135725
-0.280678 -0.097247 -0.311272 -
0.588983 -0.372357 -0.433238 0.857978
-0.591384 -1.116899 -0.252516 -1.542394 -0.097331
-0.069379 1.536376 0.933831 -1.623348 -0.496312
-0.588361 -0.203984
3 0.000006 udp - INT 2 0 900 0
166666.660800 254 0 6.000000e+08 0.000000 0 0
0.006000 0.000000 0.000000 0.000000 0 0 0.000000
0.000000 0.000000 450 0 0 0
3 2 2 2 1
3 0 0 0 2 3
0 0 udp - INT False
False False False False True False
True False False False False
False False False False False False

```
True      False        False  20.212440   0.000000    6.803505
0.000000  0.000000  0.000000  12.023757    0.005982    0.000000
1.098612   0.000000   0.000000   0.000000                    0.0
2.069854         -1.649765          -0.040322          -1.650266
-1.405884        -1.200567         1.688007          -1.188064
-1.103353         -1.217874         -1.440439         -1.199338
-0.956103                      -0.307674      -0.590835          -
0.282654                -0.217036              -0.135725
-0.280678          -0.097247                -0.058123          -
0.385295          -0.372357          -0.433238             0.857978
-0.591384    -1.116899   -0.252516    -1.542394            -0.097331
-0.069379      1.185241     0.933831    -1.623348      -0.496312
-0.588361           -0.203984
4      0.000010   udp       -   INT      2      0    2126       0
100000.002500   254     0  8.504000e+08     0.000000      0      0
0.010000    0.000000     0.000000    0.000000     0     0  0.000000
0.000000  0.000000   1063      0            0                  0
3             2           2                 2                 1
3             0           0                 0           2           3
0      0            udp                -            INT      False
False        False       False      False       True       False
True        False        False             False         False
False         False         False        False      False      False
True      False        False  20.561217   0.000000    7.662468
0.000000  0.000000  0.000000  11.512935    0.009950    0.000000
1.098612   0.000000   0.000000   0.000000                    0.0
2.159247         -1.649765           0.507665          -1.650266
-1.405884        -1.200567         1.543799          -1.186245
-1.103353         -1.217874         -1.440439         -1.199338
-0.956103                      -0.307674      -0.590835          -
0.282654                -0.217036              -0.135725
-0.280678          -0.097247                -0.058123          -
0.385295          -0.372357          -0.433238             0.857978
-0.591384    -1.116899   -0.252515    -1.542394            -0.097331
-0.069379      3.747693     0.933831    -1.623348      -0.496312
-0.588361           -0.203984
...         ...   ...     ...   ...    ...    ...     ...     ...
...   ...   ...           ...          ...    ...    ...        ...
...          ...         ...   ...   ...       ...       ...       ...
...    ...          ...                ...         ...           ...
...               ...               ...             ...           ...
...               ...         ...         ...              ...    ...
...              ...             ...        ...         ...          ..
.         ...        ...        ...         ...        ...          ..
.          ...                ...           ...            ...
...           ...          ...        ...        ...        ...
...          ...        ...        ...         ...         ...       .
..       ...        ...         ...         ...        ...        ...
...        ...                     ...                ...
```

```
...                ...                ...              ...
...              ...                ...                ...
...                ...                ...                ...
...            ...                ...                        ...
...                      ...                ...                          .
..                ...                ...                ...
...           ...          ...          ...          ...
...                     ...           ...          ...          ...
...             ...                ...
55940  0.811914   tcp       -   FIN     10      6     588     268
18.474862   254   252  5.222228e+03  2207.130127      2      1
83.216333  138.888594  4368.102356  195.623781   255   255  0.237125
0.117467  0.119658     59     45            0                  0
4             1           1                 1                 1
4             0           0                 0           1           4
0      0            tcp                -            FIN      False
False        False       False       True      False       False
True        False        False             False         False
False         False         False        False      False       True
False      False        False   8.560871   7.699901    6.378426
5.594711  8.382313  5.281292   2.969125    4.433389    4.940846
2.397895   1.945910   1.098612   0.693147                    0.0
-0.916508         -0.109794          -0.311507          -0.002247
0.836227         0.727900        -0.868175           0.841489
1.161766         -0.003422         -0.052435         -0.049192
-0.345548                     -0.307674       1.329251          -
0.503144                -0.217036               0.013658
-0.280678          -0.097247                -0.311272          -
0.588983          -0.228687          -0.291907            -0.065322
-0.429739     1.020781   -0.054004     0.648468            -0.097331
-0.069379     -0.449210     0.933831     0.616090       0.944470
1.216851           -0.203984
55941  0.000004   udp       -   INT      2      0     104       0
250000.000600   254     0  1.040000e+08     0.000000      0      0
0.004000    0.000000     0.000000    0.000000     0     0  0.000000
0.000000  0.000000     52      0            0                  0
1             2           1                 1                 1
1             0           0                 0           1           1
0      0            udp                -            INT      False
False        False       False      False       True       False
True        False        False             False         False
False         False         False        False      False      False
True      False        False  18.459901   0.000000    4.653960
0.000000  0.000000  0.000000  12.429220    0.003992    0.000000
1.098612   0.000000   0.000000   0.000000                    0.0
1.620669         -1.649765          -1.411653          -1.650266
-1.405884        -1.200567         1.802472          -1.188976
-1.103353         -1.217874         -1.440439         -1.199338
-0.956103                     -0.307674      -0.590835          -
```

0.503144 -0.217036 -0.434490
-0.280678 -0.097247 -0.311272 -
0.588983 -0.659696 -0.715899 0.857978
-0.591384 -1.116899 -0.252516 -1.542394 -0.097331
-0.069379 -0.478471 0.933831 -1.623348 -0.496312
-0.588361 -0.203984
55942 5.167410 tcp - FIN 108 24 138067 1042
25.351192 254 252 2.117719e+05 1546.616211 51 1
48.287767 221.155219 9709.118613 559.560500 255 255 0.140691
0.080831 0.059860 1278 43 0 0
1 1 1 1 1
1 0 0 0 1 3
0 0 tcp - FIN False
False False False False True False False
True False False False False
False False False False False True
False False False 12.263270 7.344471 11.835502
6.949856 9.180924 6.328937 3.271514 3.897676 5.403376
4.691348 3.218876 3.951244 0.693147 0.0
0.032438 -0.180879 3.169907 0.396934
1.049840 1.110448 -0.782808 0.595914
1.373812 2.140291 0.855562 2.937250
-0.345548 -0.307674 0.369705 -
0.503144 -0.217036 -0.434490
-0.280678 -0.097247 -0.311272 -
0.588983 -0.372357 -0.715899 -0.065322
-0.436923 1.020781 1.010917 0.648468 -0.097331
-0.069379 4.646432 0.933831 0.616090 0.495114
0.482707 -0.203984
55943 0.000005 udp - INT 2 0 104 0
200000.005100 254 0 8.320000e+07 0.000000 0 0
0.005000 0.000000 0.000000 0.000000 0 0 0.000000
0.000000 0.000000 52 0 0 0
1 2 2 1 1
2 0 0 0 2 1
0 0 udp - INT False
False False False False False True False
True False False False False
False False False False False False
True False False 18.236758 0.000000 4.653960
0.000000 0.000000 0.000000 12.206078 0.004988 0.000000
1.098612 0.000000 0.000000 0.000000 0.0
1.563476 -1.649765 -1.411653 -1.650266
-1.405884 -1.200567 1.739477 -1.188520
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.282654 -0.217036 -0.285107
-0.280678 -0.097247 -0.311272 -
0.385295 -0.659696 -0.715899 0.857978

```
-0.591384    -1.116899   -0.252516    -1.542394            -0.097331
-0.069379     -0.478471     0.933831    -1.623348      -0.496312
-0.588361           -0.203984
55944  1.106101   tcp       -   FIN     20      8   18062     354
24.410067   254   252  1.241044e+05  2242.109863      7      1
55.880051  143.700000  4798.130981  190.980813   255   255  0.173208
0.100191  0.073017    903     44            0                  0
1             1           2                 1                 1
1             0           0                 0           3           2
0      0            tcp                 -            FIN      False
False        False       False       True      False       False
True        False        False             False         False
False         False         False        False      False       True
False      False        False  11.728886   7.715619    9.801621
5.872118  8.476190  5.257395   3.235145    4.040945    4.974663
3.044522   2.197225   2.079442   0.693147                    0.0
-0.104528         -0.106650           1.872366           0.079468
0.861337         0.719174        -0.793075           0.661590
1.177269          0.600987          0.126825          0.977646
-0.345548                      -0.307674       0.580828          -
0.282654                -0.217036              -0.434490
-0.280678          -0.097247                -0.311272          -
0.181606          -0.516027          -0.715899            -0.065322
-0.433331     1.020781    0.017925     0.648468            -0.097331
-0.069379      3.078863     0.933831     0.616090       0.732573
0.730256           -0.203984

[55945 rows x 116 columns]
```

df_full: every column: originals, log-transformed, OHE, scaled, and label
df_model: for LR & MLP: OHE + log-then-scaled (skewed) + scaled (non-skewed) + label
df_model_tree: for RF & XGBoost: OHE + raw numerical columns + label (no scaling needed)

```python
df_full = df.copy()
df_model = df[ohe_cols + scaled_cols + ['label']].copy()
df_model_tree = df[ohe_cols + num_cols + ['label']].copy()

print(f"df_full: {df_full.shape}")
print(f"df_model: {df_model.shape}")
print(f"df_model_tree: {df_model_tree.shape}")
```

```
df_full: (55945, 116)
df_model: (55945, 59)
df_model_tree: (55945, 59)
```

```python
df_full
```

```
            dur proto service state  spkts  dpkts  sbytes  dbytes
rate  sttl  dttl         sload        dload  sloss  dloss     sinpkt
dinpkt         sjit        djit  swin  dwin    tcprtt    synack
```

ackdat smean dmean trans_depth response_body_len ct_srv_src
ct_state_ttl ct_dst_ltm ct_src_dport_ltm ct_dst_sport_ltm
ct_dst_src_ltm is_ftp_login ct_ftp_cmd ct_flw_http_mthd ct_src_ltm
ct_srv_dst is_sm_ips_ports label proto_filtered service_filtered
state_filtered proto_arp proto_ospf proto_other proto_sctp
proto_tcp proto_udp proto_unas service_- service_dns service_ftp
service_ftp-data service_http service_other service_pop3
service_smtp service_ssh state_CON state_FIN state_INT state_REQ
state_other sload_log dload_log sbytes_log dbytes_log sjit_log
djit_log rate_log sinpkt_log dinpkt_log spkts_log dpkts_log
sloss_log dloss_log response_body_len_log sload_log_scaled
dload_log_scaled sbytes_log_scaled dbytes_log_scaled
sjit_log_scaled djit_log_scaled rate_log_scaled sinpkt_log_scaled
dinpkt_log_scaled spkts_log_scaled dpkts_log_scaled
sloss_log_scaled dloss_log_scaled response_body_len_log_scaled
ackdat_scaled ct_dst_ltm_scaled ct_dst_sport_ltm_scaled
ct_dst_src_ltm_scaled ct_flw_http_mthd_scaled ct_ftp_cmd_scaled
ct_src_dport_ltm_scaled ct_src_ltm_scaled ct_srv_dst_scaled
ct_srv_src_scaled ct_state_ttl_scaled dmean_scaled dttl_scaled
dur_scaled dwin_scaled is_ftp_login_scaled is_sm_ips_ports_scaled
smean_scaled sttl_scaled swin_scaled synack_scaled tcprtt_scaled
trans_depth_scaled
0 0.000011 udp - INT 2 0 496 0
90909.090200 254 0 1.803636e+08 0.000000 0 0
0.011000 0.000000 0.000000 0.000000 0 0 0.000000
0.000000 0.000000 248 0 0 0
2 2 1 1 1
2 0 0 0 1 2
0 0 udp - INT False
False False False False True False
True False False False False
False False False False False False
True False False 19.010486 0.000000 6.208590
0.000000 0.000000 0.000000 11.417626 0.010940 0.000000
1.098612 0.000000 0.000000 0.000000 0.0
1.761786 -1.649765 -0.419856 -1.650266
-1.405884 -1.200567 1.516892 -1.185791
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.285107
-0.280678 -0.097247 -0.311272 -
0.588983 -0.516027 -0.574568 0.857978
-0.591384 -1.116899 -0.252515 -1.542394 -0.097331
-0.069379 0.340845 0.933831 -1.623348 -0.496312
-0.588361 -0.203984
1 0.000008 udp - INT 2 0 1762 0
125000.000300 254 0 8.810000e+08 0.000000 0 0
0.008000 0.000000 0.000000 0.000000 0 0 0.000000
0.000000 0.000000 881 0 0 0

2 2 1 1 1
2 0 0 0 1 2
0 0 udp - INT False
False False False False True False
True False False False False
False False False False False False
True False False 20.596568 0.000000 7.474772
0.000000 0.000000 0.000000 11.736077 0.007968 0.000000
1.098612 0.000000 0.000000 0.000000 0.0
2.168308 -1.649765 0.387922 -1.650266
-1.405884 -1.200567 1.606793 -1.187154
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.285107
-0.280678 -0.097247 -0.311272 -
0.588983 -0.516027 -0.574568 0.857978
-0.591384 -1.116899 -0.252515 -1.542394 -0.097331
-0.069379 2.986899 0.933831 -1.623348 -0.496312
-0.588361 -0.203984
2 0.000005 udp - INT 2 0 1068 0
200000.005100 254 0 8.544000e+08 0.000000 0 0
0.005000 0.000000 0.000000 0.000000 0 0 0.000000
0.000000 0.000000 534 0 0 0
3 2 1 1 1
3 0 0 0 1 3
0 0 udp - INT False
False False False False True False
True False False False False
False False False False False False
True False False 20.565910 0.000000 6.974479
0.000000 0.000000 0.000000 12.206078 0.004988 0.000000
1.098612 0.000000 0.000000 0.000000 0.0
2.160450 -1.649765 0.068753 -1.650266
-1.405884 -1.200567 1.739477 -1.188520
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.135725
-0.280678 -0.097247 -0.311272 -
0.588983 -0.372357 -0.433238 0.857978
-0.591384 -1.116899 -0.252516 -1.542394 -0.097331
-0.069379 1.536376 0.933831 -1.623348 -0.496312
-0.588361 -0.203984
3 0.000006 udp - INT 2 0 900 0
166666.660800 254 0 6.000000e+08 0.000000 0 0
0.006000 0.000000 0.000000 0.000000 0 0 0.000000
0.000000 0.000000 450 0 0 0
3 2 2 2 1
3 0 0 0 2 3
0 0 udp - INT False

```
False        False       False      False       True       False
True        False        False             False         False
False         False         False        False      False      False
True      False        False  20.212440   0.000000    6.803505
0.000000  0.000000  0.000000  12.023757    0.005982    0.000000
1.098612   0.000000   0.000000   0.000000                    0.0
2.069854         -1.649765          -0.040322          -1.650266
-1.405884        -1.200567         1.688007          -1.188064
-1.103353         -1.217874         -1.440439         -1.199338
-0.956103                       -0.307674      -0.590835          -
0.282654                -0.217036              -0.135725
-0.280678          -0.097247                -0.058123          -
0.385295          -0.372357          -0.433238             0.857978
-0.591384    -1.116899   -0.252516    -1.542394            -0.097331
-0.069379      1.185241     0.933831    -1.623348      -0.496312
-0.588361           -0.203984
4      0.000010   udp       -   INT      2      0    2126       0
100000.002500   254     0  8.504000e+08     0.000000      0      0
0.010000    0.000000     0.000000    0.000000     0     0  0.000000
0.000000  0.000000   1063      0            0                  0
3             2           2                 2                 1
3             0           0                 0           2           3
0      0            udp                -            INT      False
False        False       False      False       True       False
True        False        False             False         False
False         False         False        False      False      False
True      False        False  20.561217   0.000000    7.662468
0.000000  0.000000  0.000000  11.512935    0.009950    0.000000
1.098612   0.000000   0.000000   0.000000                    0.0
2.159247         -1.649765           0.507665          -1.650266
-1.405884        -1.200567         1.543799          -1.186245
-1.103353         -1.217874         -1.440439         -1.199338
-0.956103                       -0.307674      -0.590835          -
0.282654                -0.217036              -0.135725
-0.280678          -0.097247                -0.058123          -
0.385295          -0.372357          -0.433238             0.857978
-0.591384    -1.116899   -0.252515    -1.542394            -0.097331
-0.069379      3.747693     0.933831    -1.623348      -0.496312
-0.588361           -0.203984
...         ...   ...     ...   ...    ...    ...     ...     ...
...   ...   ...           ...          ...    ...    ...        ...
...          ...         ...   ...   ...       ...       ...       ...
...    ...          ...                ...         ...           ...
...                ...                ...             ...           ...
...                ...         ...         ...              ...    ...
...               ...            ...        ...         ...          ..
.         ...        ...        ...         ...        ...          ..
.          ...                ...           ...            ...
...           ...          ...        ...        ...        ...
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...          ...        ...        ...         ...         ...       .
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...        ...                    ...                ...
...                ...                ...              ...
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...            ...                ...                       ...
...                     ...                ...                        .
..                ...                ...                ...
...           ...          ...         ...          ...
...                     ...           ...          ...          ...
...             ...                  ...
55940  0.811914   tcp       -   FIN     10      6     588     268
18.474862   254   252  5.222228e+03  2207.130127      2      1
83.216333  138.888594  4368.102356  195.623781   255   255  0.237125
0.117467  0.119658     59     45            0                  0
4             1           1                 1                 1
4             0           0                 0           1           4
0      0            tcp                -            FIN      False
False        False       False       True      False       False
True        False        False             False         False
False         False         False        False      False       True
False      False        False   8.560871   7.699901    6.378426
5.594711  8.382313  5.281292   2.969125    4.433389    4.940846
2.397895   1.945910   1.098612   0.693147                    0.0
-0.916508         -0.109794          -0.311507          -0.002247
0.836227         0.727900        -0.868175           0.841489
1.161766         -0.003422         -0.052435         -0.049192
-0.345548                     -0.307674       1.329251          -
0.503144                -0.217036               0.013658
-0.280678          -0.097247                -0.311272          -
0.588983          -0.228687          -0.291907            -0.065322
-0.429739     1.020781   -0.054004     0.648468            -0.097331
-0.069379     -0.449210     0.933831     0.616090       0.944470
1.216851           -0.203984
55941  0.000004   udp       -   INT      2      0     104       0
250000.000600   254     0  1.040000e+08     0.000000      0      0
0.004000    0.000000     0.000000    0.000000     0     0  0.000000
0.000000  0.000000     52      0            0                  0
1             2           1                 1                 1
1             0           0                 0           1           1
0      0            udp                -            INT      False
False        False       False      False       True       False
True        False        False             False         False
False         False         False        False      False      False
True      False        False  18.459901   0.000000    4.653960
0.000000  0.000000  0.000000  12.429220    0.003992    0.000000
1.098612   0.000000   0.000000   0.000000                    0.0
1.620669         -1.649765          -1.411653          -1.650266
```

-1.405884 -1.200567 1.802472 -1.188976
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.434490
-0.280678 -0.097247 -0.311272 -
0.588983 -0.659696 -0.715899 0.857978
-0.591384 -1.116899 -0.252516 -1.542394 -0.097331
-0.069379 -0.478471 0.933831 -1.623348 -0.496312
-0.588361 -0.203984
55942 5.167410 tcp - FIN 108 24 138067 1042
25.351192 254 252 2.117719e+05 1546.616211 51 1
48.287767 221.155219 9709.118613 559.560500 255 255 0.140691
0.080831 0.059860 1278 43 0 0
1 1 1 1 1
1 0 0 0 1 3
0 0 tcp - FIN False
False False False True False False
True False False False False False
False False False False False True
False False False 12.263270 7.344471 11.835502
6.949856 9.180924 6.328937 3.271514 3.897676 5.403376
4.691348 3.218876 3.951244 0.693147 0.0
0.032438 -0.180879 3.169907 0.396934
1.049840 1.110448 -0.782808 0.595914
1.373812 2.140291 0.855562 2.937250
-0.345548 -0.307674 0.369705 -
0.503144 -0.217036 -0.434490
-0.280678 -0.097247 -0.311272 -
0.588983 -0.372357 -0.715899 -0.065322
-0.436923 1.020781 1.010917 0.648468 -0.097331
-0.069379 4.646432 0.933831 0.616090 0.495114
0.482707 -0.203984
55943 0.000005 udp - INT 2 0 104 0
200000.005100 254 0 8.320000e+07 0.000000 0 0
0.005000 0.000000 0.000000 0.000000 0 0 0.000000
0.000000 0.000000 52 0 0 0
1 2 2 1 1
2 0 0 0 2 1
0 0 udp - INT False
False False False False True False
True False False False False False
False False False False False False
True False False 18.236758 0.000000 4.653960
0.000000 0.000000 0.000000 12.206078 0.004988 0.000000
1.098612 0.000000 0.000000 0.000000 0.0
1.563476 -1.649765 -1.411653 -1.650266
-1.405884 -1.200567 1.739477 -1.188520
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -

```
0.282654                -0.217036              -0.285107
-0.280678          -0.097247                -0.311272          -
0.385295          -0.659696          -0.715899             0.857978
-0.591384    -1.116899   -0.252516    -1.542394            -0.097331
-0.069379     -0.478471     0.933831    -1.623348      -0.496312
-0.588361           -0.203984
55944  1.106101   tcp       -   FIN     20      8   18062     354
24.410067   254   252  1.241044e+05  2242.109863      7      1
55.880051  143.700000  4798.130981  190.980813   255   255  0.173208
0.100191  0.073017    903     44            0                  0
1             1           2                 1                 1
1             0           0                 0           3           2
0      0            tcp                -            FIN      False
False        False       False       True      False       False
True        False        False             False         False
False         False         False        False      False       True
False      False        False  11.728886   7.715619    9.801621
5.872118  8.476190  5.257395   3.235145    4.040945    4.974663
3.044522   2.197225   2.079442   0.693147                     0.0
-0.104528         -0.106650           1.872366           0.079468
0.861337         0.719174        -0.793075           0.661590
1.177269          0.600987          0.126825          0.977646
-0.345548                      -0.307674       0.580828          -
0.282654                -0.217036              -0.434490
-0.280678          -0.097247                -0.311272          -
0.181606          -0.516027          -0.715899            -0.065322
-0.433331     1.020781    0.017925     0.648468            -0.097331
-0.069379      3.078863     0.933831     0.616090       0.732573
0.730256           -0.203984

[55945 rows x 116 columns]
```

df_model

```
       proto_arp  proto_ospf  proto_other  proto_sctp  proto_tcp
proto_udp  proto_unas  service_-  service_dns  service_ftp
service_ftp-data  service_http  service_other  service_pop3
service_smtp  service_ssh  state_CON  state_FIN  state_INT  state_REQ
state_other  sload_log_scaled  dload_log_scaled  sbytes_log_scaled
dbytes_log_scaled  sjit_log_scaled  djit_log_scaled  rate_log_scaled
sinpkt_log_scaled  dinpkt_log_scaled  spkts_log_scaled
dpkts_log_scaled  sloss_log_scaled  dloss_log_scaled
response_body_len_log_scaled  ackdat_scaled  ct_dst_ltm_scaled
ct_dst_sport_ltm_scaled  ct_dst_src_ltm_scaled
ct_flw_http_mthd_scaled  ct_ftp_cmd_scaled  ct_src_dport_ltm_scaled
ct_src_ltm_scaled  ct_srv_dst_scaled  ct_srv_src_scaled
ct_state_ttl_scaled  dmean_scaled  dttl_scaled  dur_scaled
dwin_scaled  is_ftp_login_scaled  is_sm_ips_ports_scaled  smean_scaled
sttl_scaled  swin_scaled  synack_scaled  tcprtt_scaled
trans_depth_scaled  label
```

0 False False False False False
True False True False False
False False False False False
False False False True False False
1.761786 -1.649765 -0.419856 -1.650266
-1.405884 -1.200567 1.516892 -1.185791
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.285107
-0.280678 -0.097247 -0.311272 -
0.588983 -0.516027 -0.574568 0.857978
-0.591384 -1.116899 -0.252515 -1.542394 -0.097331
-0.069379 0.340845 0.933831 -1.623348 -0.496312
-0.588361 -0.203984 0
1 False False False False False
True False True False False
False False False False False
False False False True False False
2.168308 -1.649765 0.387922 -1.650266
-1.405884 -1.200567 1.606793 -1.187154
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.285107
-0.280678 -0.097247 -0.311272 -
0.588983 -0.516027 -0.574568 0.857978
-0.591384 -1.116899 -0.252515 -1.542394 -0.097331
-0.069379 2.986899 0.933831 -1.623348 -0.496312
-0.588361 -0.203984 0
2 False False False False False
True False True False False
False False False False False
False False False True False False
2.160450 -1.649765 0.068753 -1.650266
-1.405884 -1.200567 1.739477 -1.188520
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.135725
-0.280678 -0.097247 -0.311272 -
0.588983 -0.372357 -0.433238 0.857978
-0.591384 -1.116899 -0.252516 -1.542394 -0.097331
-0.069379 1.536376 0.933831 -1.623348 -0.496312
-0.588361 -0.203984 0
3 False False False False False
True False True False False
False False False False False
False False False True False False
2.069854 -1.649765 -0.040322 -1.650266
-1.405884 -1.200567 1.688007 -1.188064
-1.103353 -1.217874 -1.440439 -1.199338

```
-0.956103                      -0.307674      -0.590835          -
0.282654                -0.217036              -0.135725
-0.280678          -0.097247                -0.058123          -
0.385295          -0.372357          -0.433238             0.857978
-0.591384    -1.116899   -0.252516    -1.542394            -0.097331
-0.069379      1.185241     0.933831    -1.623348      -0.496312
-0.588361           -0.203984      0
4          False       False        False       False      False
True       False       True        False        False
False         False          False         False         False
False      False      False       True      False        False
2.159247         -1.649765           0.507665          -1.650266
-1.405884        -1.200567         1.543799          -1.186245
-1.103353         -1.217874         -1.440439         -1.199338
-0.956103                      -0.307674      -0.590835          -
0.282654                -0.217036              -0.135725
-0.280678          -0.097247                -0.058123          -
0.385295          -0.372357          -0.433238             0.857978
-0.591384    -1.116899   -0.252515    -1.542394            -0.097331
-0.069379      3.747693     0.933831    -1.623348      -0.496312
-0.588361           -0.203984      0
...          ...         ...          ...         ...        ...
...         ...        ...          ...          ...                ...
...            ...           ...           ...          ...        ...
...        ...        ...          ...                ...
...                ...                ...              ...
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...            ...                ...                       ...
...                      ...                ...                          .
..                ...                ...                ...
...            ...          ...         ...          ...
...                      ...           ...          ...          ...
...            ...                ...    ...
55940      False       False        False       False       True
False       False       True        False        False
False         False          False         False         False
False      False       True      False      False        False
-0.916508         -0.109794          -0.311507          -0.002247
0.836227         0.727900        -0.868175           0.841489
1.161766         -0.003422         -0.052435         -0.049192
-0.345548                      -0.307674       1.329251          -
0.503144                -0.217036               0.013658
-0.280678          -0.097247                -0.311272          -
0.588983          -0.228687          -0.291907            -0.065322
-0.429739     1.020781   -0.054004     0.648468            -0.097331
-0.069379     -0.449210     0.933831     0.616090       0.944470
1.216851           -0.203984      0
55941      False       False        False       False      False
```

True False True False False
False False False False False
False False False True False False
1.620669 -1.649765 -1.411653 -1.650266
-1.405884 -1.200567 1.802472 -1.188976
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.503144 -0.217036 -0.434490
-0.280678 -0.097247 -0.311272 -
0.588983 -0.659696 -0.715899 0.857978
-0.591384 -1.116899 -0.252516 -1.542394 -0.097331
-0.069379 -0.478471 0.933831 -1.623348 -0.496312
-0.588361 -0.203984 0
55942 False False False False True
False False True False False
False False False False False
False False True False False False
0.032438 -0.180879 3.169907 0.396934
1.049840 1.110448 -0.782808 0.595914
1.373812 2.140291 0.855562 2.937250
-0.345548 -0.307674 0.369705 -
0.503144 -0.217036 -0.434490
-0.280678 -0.097247 -0.311272 -
0.588983 -0.372357 -0.715899 -0.065322
-0.436923 1.020781 1.010917 0.648468 -0.097331
-0.069379 4.646432 0.933831 0.616090 0.495114
0.482707 -0.203984 0
55943 False False False False False
True False True False False
False False False False False
False False False True False False
1.563476 -1.649765 -1.411653 -1.650266
-1.405884 -1.200567 1.739477 -1.188520
-1.103353 -1.217874 -1.440439 -1.199338
-0.956103 -0.307674 -0.590835 -
0.282654 -0.217036 -0.285107
-0.280678 -0.097247 -0.311272 -
0.385295 -0.659696 -0.715899 0.857978
-0.591384 -1.116899 -0.252516 -1.542394 -0.097331
-0.069379 -0.478471 0.933831 -1.623348 -0.496312
-0.588361 -0.203984 0
55944 False False False False True
False False True False False
False False False False False
False False True False False False
-0.104528 -0.106650 1.872366 0.079468
0.861337 0.719174 -0.793075 0.661590
1.177269 0.600987 0.126825 0.977646
-0.345548 -0.307674 0.580828 -

```
0.282654                -0.217036              -0.434490
-0.280678          -0.097247                -0.311272          -
0.181606          -0.516027          -0.715899            -0.065322
-0.433331     1.020781    0.017925     0.648468            -0.097331
-0.069379      3.078863     0.933831     0.616090       0.732573
0.730256           -0.203984      0

[55945 rows x 59 columns]
```

```
df_model.columns
```

```
Index(['proto_arp', 'proto_ospf', 'proto_other', 'proto_sctp',
'proto_tcp',
       'proto_udp', 'proto_unas', 'service_-', 'service_dns',
'service_ftp',
       'service_ftp-data', 'service_http', 'service_other',
'service_pop3',
       'service_smtp', 'service_ssh', 'state_CON', 'state_FIN',
'state_INT',
       'state_REQ', 'state_other', 'sload_log_scaled',
'dload_log_scaled',
       'sbytes_log_scaled', 'dbytes_log_scaled', 'sjit_log_scaled',
       'djit_log_scaled', 'rate_log_scaled', 'sinpkt_log_scaled',
       'dinpkt_log_scaled', 'spkts_log_scaled', 'dpkts_log_scaled',
       'sloss_log_scaled', 'dloss_log_scaled',
'response_body_len_log_scaled',
       'ackdat_scaled', 'ct_dst_ltm_scaled',
'ct_dst_sport_ltm_scaled',
       'ct_dst_src_ltm_scaled', 'ct_flw_http_mthd_scaled',
'ct_ftp_cmd_scaled',
       'ct_src_dport_ltm_scaled', 'ct_src_ltm_scaled',
'ct_srv_dst_scaled',
       'ct_srv_src_scaled', 'ct_state_ttl_scaled', 'dmean_scaled',
       'dttl_scaled', 'dur_scaled', 'dwin_scaled',
'is_ftp_login_scaled',
       'is_sm_ips_ports_scaled', 'smean_scaled', 'sttl_scaled',
'swin_scaled',
       'synack_scaled', 'tcprtt_scaled', 'trans_depth_scaled',
'label'],
      dtype='str')
```

```
df_model_tree
```

```
       proto_arp  proto_ospf  proto_other  proto_sctp  proto_tcp
proto_udp  proto_unas  service_-  service_dns  service_ftp
service_ftp-data  service_http  service_other  service_pop3
service_smtp  service_ssh  state_CON  state_FIN  state_INT  state_REQ
state_other    ackdat  ct_dst_ltm  ct_dst_sport_ltm  ct_dst_src_ltm
ct_flw_http_mthd  ct_ftp_cmd  ct_src_dport_ltm  ct_src_ltm  ct_srv_dst
ct_srv_src  ct_state_ttl  dbytes      dinpkt        djit        dload
```

```
dloss  dmean  dpkts  dttl       dur  dwin  is_ftp_login
is_sm_ips_ports           rate  response_body_len  sbytes     sinpkt
sjit         sload  sloss  smean  spkts  sttl  swin    synack
tcprtt  trans_depth  label
0          False       False        False       False      False
True       False       True        False        False
False         False          False         False         False
False      False      False       True      False        False
0.000000           1                  1               2
0           0                  1           1           2           2
2       0    0.000000    0.000000     0.000000      0      0      0
0  0.000011     0             0                 0   90909.090200
0     496   0.011000     0.000000  1.803636e+08      0    248      2
254     0  0.000000  0.000000            0      0
1          False       False        False       False      False
True       False       True        False        False
False         False          False         False         False
False      False      False       True      False        False
0.000000           1                  1               2
0           0                  1           1           2           2
2       0    0.000000    0.000000     0.000000      0      0      0
0  0.000008     0             0                 0  125000.000300
0    1762   0.008000     0.000000  8.810000e+08      0    881      2
254     0  0.000000  0.000000            0      0
2          False       False        False       False      False
True       False       True        False        False
False         False          False         False         False
False      False      False       True      False        False
0.000000           1                  1               3
0           0                  1           1           3           3
2       0    0.000000    0.000000     0.000000      0      0      0
0  0.000005     0             0                 0  200000.005100
0    1068   0.005000     0.000000  8.544000e+08      0    534      2
254     0  0.000000  0.000000            0      0
3          False       False        False       False      False
True       False       True        False        False
False         False          False         False         False
False      False      False       True      False        False
0.000000           2                  1               3
0           0                  2           2           3           3
2       0    0.000000    0.000000     0.000000      0      0      0
0  0.000006     0             0                 0  166666.660800
0     900   0.006000     0.000000  6.000000e+08      0    450      2
254     0  0.000000  0.000000            0      0
4          False       False        False       False      False
True       False       True        False        False
False         False          False         False         False
False      False      False       True      False        False
0.000000           2                  1               3
```

0 0 2 2 3 3
2 0 0.000000 0.000000 0.000000 0 0 0
0 0.000010 0 0 0 100000.002500
0 2126 0.010000 0.000000 8.504000e+08 0 1063 2
254 0 0.000000 0.000000 0 0
... ...
... ...
... ...
... ...
... ...
... ...
... ...
... ...
... ...
... ...
55940 False False False False True
False False True False False
False False False False False
False False True False False False
0.119658 1 1 4
0 0 1 1 4 4
1 268 138.888594 195.623781 2207.130127 1 45 6
252 0.811914 255 0 0 18.474862
0 588 83.216333 4368.102356 5.222228e+03 2 59 10
254 255 0.117467 0.237125 0 0
55941 False False False False False
True False True False False
False False False False False
False False False True False False
0.000000 1 1 1
0 0 1 1 1 1
2 0 0.000000 0.000000 0.000000 0 0 0
0 0.000004 0 0 0 250000.000600
0 104 0.004000 0.000000 1.040000e+08 0 52 2
254 0 0.000000 0.000000 0 0
55942 False False False False True
False False True False False
False False False False False
False False True False False False
0.059860 1 1 1
0 0 1 1 3 1
1 1042 221.155219 559.560500 1546.616211 1 43 24
252 5.167410 255 0 0 25.351192
0 138067 48.287767 9709.118613 2.117719e+05 51 1278 108
254 255 0.080831 0.140691 0 0
55943 False False False False False
True False True False False
False False False False False
False False False True False False

```
0.000000           2                  1                2
0           0                  1           2           1           1
2       0    0.000000    0.000000     0.000000      0      0      0
0  0.000005     0             0                 0  200000.005100
0     104   0.005000     0.000000  8.320000e+07      0     52      2
254     0  0.000000  0.000000            0      0
55944      False       False        False       False       True
False       False       True        False        False
False         False          False         False         False
False      False       True      False      False        False
0.073017           2                  1                1
0           0                  1           3           2           1
1     354  143.700000  190.980813  2242.109863      1     44      8
252  1.106101   255             0                 0      24.410067
0   18062  55.880051  4798.130981  1.241044e+05      7    903     20
254   255  0.100191  0.173208            0      0

[55945 rows x 59 columns]
```

```
df_model_tree.columns
```

```
Index(['proto_arp', 'proto_ospf', 'proto_other', 'proto_sctp',
'proto_tcp',
       'proto_udp', 'proto_unas', 'service_-', 'service_dns',
'service_ftp',
       'service_ftp-data', 'service_http', 'service_other',
'service_pop3',
       'service_smtp', 'service_ssh', 'state_CON', 'state_FIN',
'state_INT',
       'state_REQ', 'state_other', 'ackdat', 'ct_dst_ltm',
'ct_dst_sport_ltm',
       'ct_dst_src_ltm', 'ct_flw_http_mthd', 'ct_ftp_cmd',
'ct_src_dport_ltm',
       'ct_src_ltm', 'ct_srv_dst', 'ct_srv_src', 'ct_state_ttl',
'dbytes',
       'dinpkt', 'djit', 'dload', 'dloss', 'dmean', 'dpkts', 'dttl',
'dur',
       'dwin', 'is_ftp_login', 'is_sm_ips_ports', 'rate',
'response_body_len',
       'sbytes', 'sinpkt', 'sjit', 'sload', 'sloss', 'smean', 'spkts',
'sttl',
       'swin', 'synack', 'tcprtt', 'trans_depth', 'label'],
      dtype='str')
```

```
df_full.to_csv("../data/processed/training_full.csv", index=False)
df_model.to_csv("../data/processed/training_lr_mlp.csv", index=False)
df_model_tree.to_csv("../data/processed/training_xgb_rf.csv",
index=False)
```

Preprocessing Pipeline

Using preprocessing.py

```
df_full = pd.read_csv("../data/processed/training_full.csv")
df_model = pd.read_csv("../data/processed/training_lr_mlp.csv")
df_model_tree = pd.read_csv("../data/processed/training_xgb_rf.csv")

import sys
sys.path.append('../src')

from preprocessing.preprocessing import preprocess_train,
preprocess_test, save_artifacts

raw_train = pd.read_csv('../data/raw/UNSW_NB15_training-set.csv')
raw_test  = pd.read_csv('../data/raw/UNSW_NB15_testing-set.csv')

fn_full, fn_model, fn_model_tree, artifacts =
preprocess_train(raw_train)

print(f"fn_full       shape: {fn_full.shape}")
print(f"fn_model      shape: {fn_model.shape}")
print(f"fn_model_tree shape: {fn_model_tree.shape}")
```

```
fn_full       shape: (55945, 116)
fn_model      shape: (55945, 59)
fn_model_tree shape: (55945, 59)
```

```
save_artifacts(artifacts, '../data/processed/artifacts.pkl')

test_full, test_model, test_model_tree = preprocess_test(raw_test,
artifacts)

test_full.to_csv('../data/processed/test_full.csv', index=False)
test_model.to_csv('../data/processed/test_lr_mlp.csv', index=False)
test_model_tree.to_csv('../data/processed/test_xgb_rf.csv',
index=False)

print(f"test_full       shape: {test_full.shape}")
print(f"test_model      shape: {test_model.shape}")
print(f"test_model_tree shape: {test_model_tree.shape}")
```

```
Artifacts saved to ../data/processed/artifacts.pkl
test_full       shape: (107740, 116)
test_model      shape: (107740, 59)
test_model_tree shape: (107740, 59)
```